

Question 1:

A data Scientist is told that the characteristics of equation of a data Matrix has the property that the roots are all distinct and real.

a) The matrix is bound to be invertible

b) also diagonalizable

Is the data scientist assumption correct, prove it.

Answer 1:

Question 2:

Let $A = m \times n$ matrix over $\mathbb{R}^2$

Define $N_A = \{x \in \mathbb{R}^n : Ax = 0\}$

$N_A = \text{kernel of } A$

$v(A) = \text{dimension}(N_A)$ as nullity of A

Find $N_A + v(A)$ in both
Example

a)

$$A = \begin{pmatrix} 1 & 2 & -1 \\ 2 & -1 & 3 \end{pmatrix},$$

$N_A = \text{kernel of } A$

2. b)

$$A = \begin{pmatrix} 2 & -1 & 3 \\ 4 & -2 & 6 \\ 6 & 3 & -9 \end{pmatrix}$$

Q3

a) $m \times m$ matrix $A = PDP^T$ where $P^{-1} = P^T$

$$\text{and } D = \begin{bmatrix} \lambda & 0 & \dots & 0 \\ 0 & \lambda^2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \lambda^m \end{bmatrix} \text{ and } |\lambda| \neq 0, 1 \text{ the find}$$

a best rank 1 approximation of A^2 with

justification

$$b) \text{ vectors } v_1 = \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix}^T$$

$$v_2 = \begin{bmatrix} 1/\sqrt{2} & -1/\sqrt{2} \end{bmatrix}^T$$

find a 2×2 matrix A such that

$$Av_i = -v_i \quad \forall i = 1, 2$$

Q4:

$v, w \in \mathbb{R}^{n \times 1}$ be non zero vectors

Identity matrix $I_n \in \mathbb{R}^{n \times n}$

a) prove matrix $B = I_n - 2(vv^T)$ is an orthogonal matrix for any v . also prove B is a symmetry matrix for any v

I = is an identity matrix

Q4 b)

Euclidean norm of $w=1$

consider the matrix $C = I_n - 2(w w^T)$

prove or disapprove whether this matrix C is an orthogonal matrix

Q4, c)

Show that when the Euclidean norm of v is equal to 1, the rank of $I_n - v v^T$ is less than n

Q5:

profit of software company on a project

is given by $x_1^3 + x_2^3 + 2x_1^2 + 4x_2^2 + 6$

x_1 = salary of employee

x_2 = general expenditure

The objective is to find Hessian matrix and whether it is positive definite or negative definite at a given stationary point, and the values of x_1 and x_2 to maximize the profit

$$x_1^3 + 2x_1^2 + x_2^3 + 4x_2^2 + 6$$

Q6:

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{bmatrix} \in \mathbb{R}^{2 \times 3}$$

Consider $f: \mathbb{R}^{2 \times 3} \rightarrow \mathbb{R}^{3 \times 3}$ given by $f(A) = A^T A = k$

Calculate gradient of k with respect to A in

matrix form